

CHAPTER

10

Resonance

10.1 INTRODUCTION

Any system having at least a pair of complex conjugate poles has a natural frequency of oscillation. If the frequency of the system driving force coincides with the natural frequency of oscillation, the system resonates and the system response becomes maximum. This phenomenon is known as '*resonance*' and the frequency at which this phenomenon occurs is known as '*resonant frequency*'.

In electrical systems, resonance occurs when the system contains at least one inductor and one capacitor. In this system, the phenomenon of cancellation of reactances when inductor and capacitor are in series or cancellation of susceptances when they are in parallel, is termed as *resonance*. The circuit under resonance is purely resistive in nature and is termed as '*resonant circuit*' or '*tuned circuit*'.

In this chapter, we consider electrical resonance in details. Electrical resonance is broadly classified into two categories:

- (a) Series Resonance, and
- (b) Parallel Resonance.

10.2 SERIES RESONANCE OR VOLTAGE RESONANCE

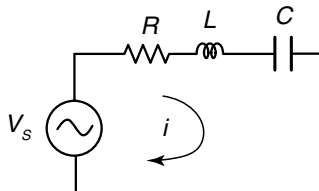


Figure 10.1 Series RLC Resonant Circuit

The basic series-resonant circuit is shown in Fig. 10.1. We want to observe how the steady state amplitude and the phase angle of the current vary with the frequency of the sinusoidal voltage source. As the frequency of the source changes, the maximum amplitude of the source voltage (V_m) is held constant.

Impedance due to capacitor: $Z_C = -\frac{j}{\omega C}$; clearly, as $\omega \rightarrow 0$, $Z_C \rightarrow \infty$ and $i \rightarrow 0$.

Impedance due to inductor: $Z_L = j\omega L$ clearly, as $\omega \rightarrow \infty$, $Z_L \rightarrow \infty$ and $i \rightarrow 0$.

Therefore, circuits containing inductors and capacitors have responses that are frequency dependent. We analyze in the following steps.

i. Current Response

Here, the supply voltage, $v_s = V_m \sin \omega t$ and the current is, $i = I_m \sin (\omega t + \phi)$. The phasor equivalents of v_s and i are $\mathbf{V}$ and $\mathbf{I}$, respectively.

$$\text{Using phasors, } I = \frac{V}{Z} = \frac{V}{R + j\omega L - \frac{j}{\omega C}} = \frac{V}{R + j\left(\omega L - \frac{1}{\omega C}\right)} \quad (10.1)$$

$$\text{Thus, the current magnitude, } |I| = \frac{V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \quad (10.2)$$

From this equation, we have the following observations:

- $|I| \rightarrow 0$ as $\omega \rightarrow 0$; and
- $|I| \rightarrow 0$ as $\omega \rightarrow \infty$.

This indicates that there must be a maximum value of the current $|I|$ for some particular value of ω . This occurs when the denominator is a minimum. i.e., when

$$\omega L = \frac{1}{\omega C} \Rightarrow \omega = \frac{1}{\sqrt{LC}} \quad (10.3)$$

Thus, resonance occurs when the magnitudes of the inductive and capacitive reactances are equal. This frequency is termed as the *resonant frequency*, ω_0 of the series *RLC* circuit.

$$\therefore \omega_0 = \frac{1}{\sqrt{LC}}$$

From equation (10.1), the phase angle of the current is given by,

$$\phi = -\tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right) \quad (10.4)$$

From equation (10.4), it is clear that the phase angle for the current also depends on frequency. We have two observations:

- As $\omega \rightarrow 0$, $\phi \rightarrow \tan^{-1}\left(\frac{1}{\omega RC}\right)$ and in this case the current leads the voltage with the phase relationship being like that of an RC circuit.
- As $\omega \rightarrow \infty$, $\phi \rightarrow -\tan^{-1}\left(\frac{\omega L}{R}\right)$ and in this case the current lags the voltage with the phase relationship being like that of an RL circuit.
- At $\omega = \omega_0$, $\phi = 0$, and in this case the current and the voltage are in phase, the circuit behaving like a purely resistive circuit.

The current response and phasor diagrams are shown in Fig. 10.2.

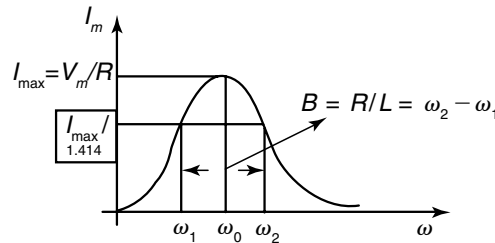


Figure 10.2 Frequency Response of a Series – Resonant Circuit

Phasor Diagrams The current and voltage phasor diagrams for an RLC series circuit are shown in Fig. 10.3.

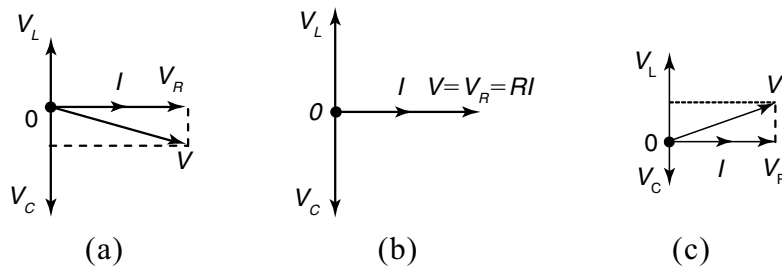


Figure 10.3 Phasor Diagrams (a) $f < f_0$ (b) $f = f_0$ (c) $f > f_0$

At resonant frequency, the inductive and capacitive reactances are equal so that the current and voltage are in phase. For any frequency lower than the resonant frequency, the inductive reactance is less than the capacitive reactance and hence, the circuit behaves as a capacitive circuit. Similarly, for any frequency higher than the resonant frequency, the inductive reactance is greater than the capacitive reactance and hence, the circuit behaves as an inductive circuit.

ii. Bandwidth

We define the half-power bandwidth of the RLC circuit as the range of frequencies (or the width of the frequency band) for which the power dissipated in R is greater than or equal to half the maximum power.

We know that the average power is,

$$P = |I|^2 R \quad \text{where} \quad |I| = \frac{I_m}{\sqrt{2}} \quad (10.5)$$

Maximum power will be, $P_{\max} = I_m^2 R$

Thus, the half-power points occur when

$$P = \frac{P_{\max}}{2} \quad \text{or} \quad |I| = \frac{I_m}{\sqrt{2}}$$

At resonance, the circuit is purely resistive, so that

$$I_m = \frac{V_m}{R} \quad (10.6)$$

Therefore, at half-power points,

$$|I| = \frac{I_m}{\sqrt{2}} = \frac{V_m}{\sqrt{2}R} \quad (10.7)$$

From equation (10.2) and (10.7), we get,

$$\frac{V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} = \frac{V_m}{\sqrt{2}R}$$

To solve for the frequencies, squaring both sides and equating the denominators,

$$R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2$$

$$\Rightarrow \left(\omega L - \frac{1}{\omega C}\right)^2 = R^2$$

$$\Rightarrow \left(\omega L - \frac{1}{\omega C}\right) = \pm R$$

$$\text{Therefore,} \quad \omega = \pm \frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

We see that, mathematically, there are 4 possible values of ω . Taking the positive roots, the half-power frequencies are:

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \quad (10.8)$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \quad (10.9)$$

By definition, the Bandwidth (BW) is given by,

$$BW = \omega_2 - \omega_1 = \frac{R}{L} \quad (10.10)$$

Also, from equations (10.8) and (10.9), we get

$$\omega_1 \omega_2 = \frac{1}{LC} = \omega_0^2 \Rightarrow \omega_0 = \sqrt{\omega_1 \omega_2} \quad (10.11)$$

Thus, the resonant frequency is the geometric mean of the half-power frequencies.

iii. Quality Factor or Circuit Magnification Factor (Q)

It is defined as, $Q = 2\pi \frac{\text{Maximum Energy Stored}}{\text{Energy Dissipated per cycle}}$

Maximum Energy stored = Electromagnetic energy in inductor or Electrostatic energy in capacitor

$$= \frac{1}{2} L I_{\max}^2 \text{ or } \frac{1}{2} C V_{\max}^2$$

Therefore,

$$Q = 2\pi \frac{\frac{1}{2} L I_{\max}^2}{\left(\frac{I_{\max}}{\sqrt{2}} \right)^2 R \times \frac{1}{f}} = \frac{2\pi f L}{R} = \frac{\omega L}{R}$$

or,
$$Q = \frac{\omega L}{R} = \frac{1}{\omega RC} = \frac{1}{R} \sqrt{\frac{L}{C}} \quad (10.12)$$

Thus, Q is inversely proportional to R . Hence, for series RLC circuit, a high value of quality factor implies low losses and a low value of Q implies high losses.

Also, *quality factor* for a circuit is defined as, $Q = \frac{\text{Resonant Frequency}}{\text{Bandwidth}} = \frac{\omega_0}{\omega_2 - \omega_1}$ and the *selectivity* of the circuit is defined as the reciprocal of quality factor, i.e.

$$\text{Selectivity} = \frac{1}{Q} = \frac{BW}{\omega_0}$$

Therefore, a circuit will be highly selective if it has a high value of Q . For series RLC circuit, a high value of quality factor implies a narrow resonant peak and a low value of Q implies broad resonant peak. The variations of magnitude and phase angle of current in RLC series circuit for different values of Quality factor (Q) are shown in Fig. 10.4.

i. Voltage Across Elements

Voltage across Resistance

Since $V = RI$ and at resonance $V_R = V_m$,

Therefore,
$$I = \frac{V_m}{R} \quad (10.13)$$

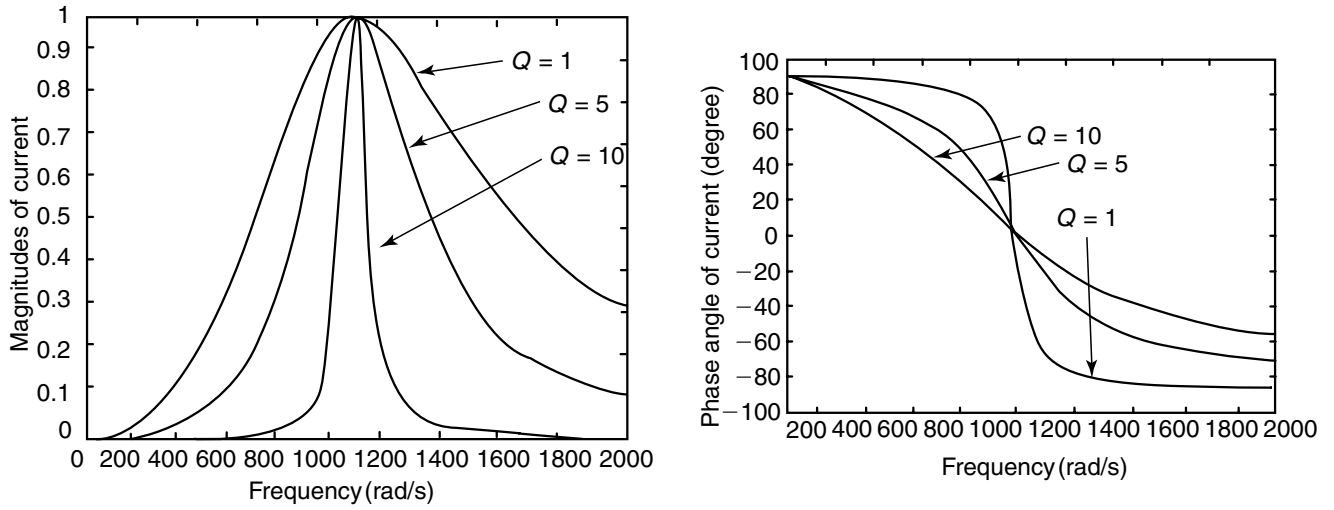


Figure 10.4 Variation of Magnitude and Phase Angle of Current in RLC Series Circuit for Different Values of Q

Voltage across Inductance

We know, $V_L = IZ_L = j\omega LI$, using equation (10.2), we get,

$$|V_L| = \omega L |I| = \frac{\omega L V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$\text{At resonance, (i.e. } \omega_0 L = \frac{1}{\omega_0 C} \text{), } |V_L| = \frac{\omega_0 L V_m}{R} = Q V_m \quad (10.14)$$

To find the frequency at which inductor voltage will be maximum, we have,

$$\begin{aligned} \frac{d}{d\omega} [|V_L|] &= 0 \\ \Rightarrow \frac{d}{d\omega} \left[\frac{\omega L V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \right] &= 0 \\ \Rightarrow \omega_L &= \frac{1}{\sqrt{LC - \frac{C^2 R^2}{2}}} \end{aligned}$$

$$\text{Thus, } f_L = \left(\frac{1}{2\pi} \right) \left(\frac{1}{\sqrt{LC - \frac{C^2 R^2}{2}}} \right) \quad (10.15)$$

Voltage across Capacitance

We know, $V_C = \frac{I}{j\omega C}$; using equation (10.2), we get,

$$|V_C| = \frac{|I|}{\omega C} = \frac{V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \frac{1}{\omega C}$$

At resonance, (i.e. $\omega_0 L = \frac{1}{\omega_0 C}$),

$$|V_C| = \frac{V_m}{\omega_0 R C} = Q V_m \quad (10.16)$$

To find the frequency at which capacitor voltage will be maximum, we have,

$$\frac{d}{d\omega} [|V_C|] = 0$$

$$\Rightarrow \frac{d}{d\omega} \left[\frac{V_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \frac{1}{\omega C} \right] = 0$$

$$\Rightarrow \omega_C = \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}}$$

$$\text{Thus } f_C = \left(\frac{1}{2\pi} \right) \sqrt{\left(\frac{1}{LC} - \frac{R^2}{2L^2} \right)} \quad (10.17)$$

From equations (10.14) and (10.16) we see that both V_L and V_C may be very large at resonance and they will add to zero (voltage across L and C are 180° out of phase). At resonant condition, the voltage across the inductor and capacitor are equal in magnitude and opposite in sign, thus canceling each other so that the entire voltage appears across the resistance. For this reason, the resonance in series RLC circuit is known as *voltage resonance*. Also, at resonant condition, from equations (10.14) and (10.16) we get,

$$|V_C| = |V_L| = Q V_m \Rightarrow Q = \frac{|V_C|}{V_m} = \frac{|V_L|}{V_m}$$

Thus, quality factor for series circuit is also defined as the ratio of voltage across the inductor or capacitor at resonance to the supply voltage. For this reason, Q is also known as *circuit magnification factor* (here, voltage magnification).

From equations (10.15) and (10.17), it is observed that $f_L > f_C$. The variations of voltage across resistor, capacitor and inductor are shown in Fig. 10.5.

However, from equations (10.15) and (10.17), it is observed that if R is very small (or Q is very large), both f_L and f_C approach f_0 . For circuits with $Q \geq 10$, the maximum voltages across R , L and C will practically occur at resonant frequency f_0 .

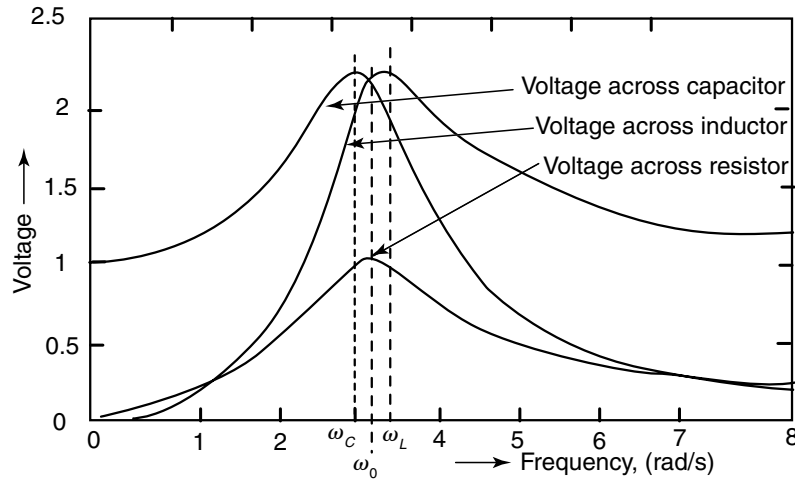


Figure 10.5 Variation of Voltage Across Resistor, Capacitor and Inductor with Frequency

Variation of Impedance with Frequency

The variations of the impedances with frequency are shown in Fig. 10.6.

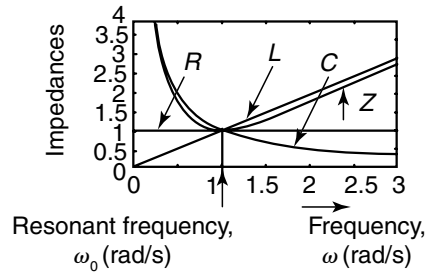


Figure 10.6 Variations of Impedances with Frequency

Here,
$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

i Impedance at Frequencies near Resonant Frequency

We introduce a new term, *fractional frequency deviation* or *fractional detuning*, δ , defined as,

$$\delta = \frac{\omega - \omega_0}{\omega_0} \Rightarrow \frac{\omega}{\omega_0} = (1 + \delta)$$

Thus, the impedance of the RLC series circuit at any frequency is given by,

$$\begin{aligned} z &= R + j\left(\omega L - \frac{1}{\omega C}\right) = R\left[1 + j\left(\frac{\omega L}{R} - \frac{1}{\omega RC}\right)\right] \\ &= R\left[1 + j\left(\frac{\omega_0 L}{R} \frac{\omega}{\omega_0} - \frac{1}{\omega_0 RC} \frac{\omega_0}{\omega}\right)\right] \end{aligned}$$

At frequencies near the resonant frequency, $\delta \ll 1$. $\therefore (2 + \delta) \approx 2$ and $(1 + \delta) \approx 1$.
 $\therefore$ Impedance near resonant frequency,

$$Z = R(1 + j2Q\delta)$$

Note: (i) At $\omega = \omega_2$ (half-power frequency) $2Q\delta = 1$; $Z = R(1 + j)$

(ii) At $\omega = \omega_1$ (half-power frequency) $2Q\delta = -1$; $Z = R(1 - j)$

Example 10.1 A series RLC circuit consists of a resistance of $1 \text{ k}\Omega$, an inductance of 10 mH and a capacitance of $100 \text{ }\mu\text{F}$. For a supply voltage of 100 V , determine the followings:

- resonant frequency,
- maximum current in the circuit,
- Q factor of the circuit, and
- Half-power frequencies.

Sol:

Here, $R = 1 \text{ k}\Omega$, $L = 10 \text{ mH}$ and $C = 100 \text{ }\mu\text{F}$, $V = 100 \text{ V}$

- (a) Resonant frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10 \times 10^{-3} \times 100 \times 10^{-6}}} = 159.15 \text{ Hz} \quad \text{Ans}$$

- (b) Maximum current in the circuit,

$$I_0 = \frac{V}{R} = \frac{100}{1 \times 10^3} = 0.1 \text{ A} \quad \text{Ans}$$

- (c) Q factor of the circuit,

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{1 \times 10^3} \sqrt{\frac{10 \times 10^{-3}}{100 \times 10^{-6}}} = 0.01 \quad \text{Ans}$$

- (d) To find half-power frequencies, we have,

$$(f_2 - f_1) = \frac{R}{4\pi L} = \frac{1 \times 10^3}{4\pi \times 10 \times 10^{-3}} = 7957.75 \quad \text{(i)}$$

$$\text{ans} \quad f_1 f_2 = \frac{1}{2\pi LC} = \frac{1}{2\pi \times 10 \times 10^{-3} \times 100 \times 10^{-6}} = 159154.94$$

$$\therefore (f_2 + f_1) = \sqrt{(f_2 - f_1)^2 + 4f_1 f_2} = \sqrt{(7957.75)^2 + 4 \times 159154.94} = 7997.65 \quad \text{(ii)}$$

Adding equations (i) + (ii), $f_2 = 7977.7 \text{ Hz}$ *Ans*

Subtracting equations (ii) - (i), $f_1 = 19.95 \text{ Hz}$ *Ans*

Example 10.2 A coil of resistance $2.2 \text{ }\Omega$ and inductance 0.01 H is connected in series with a capacitor across 220 V mains. Find the value of capacitance such that the maximum current flows in the circuit at a frequency of 100 Hz . Also, find the current and voltage across the capacitor.

Sol:

Here, $R = 2.2 \Omega$, $L = 0.01 \text{ H}$ and $V = 220 \text{ V}$, $f_0 = 100 \text{ Hz}$

$$\therefore f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{2\pi f_0^2 L} = \frac{1}{2\pi \times 100^2 \times 0.01} = 250 \text{ nF} \quad \text{Ans}$$

Current through the capacitor is the current flowing through the circuit at resonance, i.e.

$$I_C = I_0 = \frac{V}{R} = \frac{220}{2.2} = 100 \text{ A} \quad \text{Ans}$$

Voltage across the capacitor,

$$V_C = I_0 \times \frac{j}{2\pi fC} = 100 \times \frac{j}{2\pi \times 100 \times 250 \times 10^{-6}} = 636.6 \angle 90^\circ (\text{V}) \quad \text{Ans}$$

Example 10.3

A series circuit is in resonance at $8 \times 10^6 \text{ Hz}$ and has a coil of 35 mH and 10Ω resistor. For an applied voltage of 100 V ,

- Find the current at resonance,
- Find the value of the capacitance for resonance.
- Find the impedance at a frequency of 8.1 MHz and also the current at this frequency.

Sol:

Here, $R = 10 \Omega$, $L = 35 \mu\text{H}$ and $V = 100 \text{ V}$, $f_0 = 8 \times 10^6 \text{ Hz}$

- Current at resonance,

$$(b) \therefore f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{2\pi f_0^2 L} = \frac{1}{2\pi \times (8.1 \times 10^6)^2 \times 35 \times 10^{-6}} = 11.3 \text{ pF} \quad \text{Ans}$$

- To find impedance at 8.1 MHz (i.e. near resonant frequency), we have the expression as,

$$Z = R \left[1 + jQ\delta \left(\frac{2 + \delta}{1 + \delta} \right) \right]$$

$$\text{Here, } Q = \frac{\omega_0 L}{R} = \frac{2\pi \times 8 \times 10^6 \times 35 \times 10^{-6}}{10} = 176$$

$$\delta = \frac{f - f_0}{f_0} = \frac{8.1 - 8}{8} = 0.0125$$

$$\therefore Z = R \left[1 + jQ\delta \left(\frac{2 + \delta}{1 + \delta} \right) \right] = 10 \left[1 + j176 \times 0.0125 \times \left(\frac{2 + 0.0125}{1 + 0.0125} \right) \right] \\ = (10 + j43.71) \Omega \quad \text{Ans}$$

$$\therefore \text{Current at this frequency is, } I = \frac{V}{Z} = \frac{100}{(10 + j43.71)} = 2.23 \angle -77.11^\circ (\text{A}) \quad \text{Ans}$$

10.3 PARALLEL RESONANCE OR CURRENT RESONANCE OR ANTI-RESONANCE

The basic parallel-resonant circuit is shown in Fig.10.7. We want to observe how the steady state amplitude and the phase angle of the voltage vary with the frequency of the sinusoidal current source. As the frequency of the source changes, the maximum amplitude of the source current (I_m) is held constant.

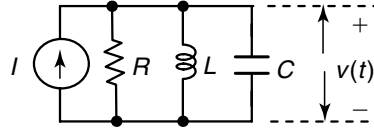


Figure 10.7 Parallel RLC Resonant Circuit

In this circuit, the impedance is given by,

$$Z = \frac{1}{\frac{1}{R} + \frac{1}{j\omega L} + j\omega C} \quad (10.18)$$

i. Voltage Response

Here, the supply current, $i_s = I_m \sin \omega t$ and the voltage is, $v = V_m \sin (\omega t + \phi)$. The phasor equivalents of i_s and v are $\mathbf{I}$ and $\mathbf{V}$, respectively.

Using phasors, $|\mathbf{V}| = IZ = \frac{I_m}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}} \quad (10.33)$

From this equation, we have the following observations:

- $|\mathbf{V}| \rightarrow 0$ as $\omega \rightarrow 0$; and
- $|\mathbf{V}| \rightarrow 0$ as $\omega \rightarrow \infty$.

This indicates that there must be a maximum value of the voltage $|\mathbf{V}|$ for some particular value of ω . This occurs when the denominator is a minimum. i.e., when

$$\omega L = \frac{1}{\omega C} \Rightarrow \omega = \frac{1}{\sqrt{LC}} \quad (10.20)$$

Thus, resonance occurs when the magnitudes of the inductive and capacitive reactances are equal. This frequency is termed as the *resonant frequency*, ω_0 of the parallel *RLC* circuit.

$$\therefore \boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

From equation (10.19), the phase angle of the voltage is given by,

$$\phi = \tan^{-1} \left(\left(\frac{1}{\omega L} - \omega C \right) R \right) \quad (10.21)$$

From equation (10.21), it is clear that the phase angle for the voltage also depends on frequency. We have two observations:

- As $\omega \rightarrow 0$, $\phi \rightarrow 90^\circ$ and in this case the current lags the voltage with the phase relationship being like that of an RL circuit.
- As $\omega \rightarrow \infty$, $\phi \rightarrow -90^\circ$ and in this case the current leads the voltage with the phase relationship being like that of an RC circuit.
- At $\omega = \omega_0$, $\phi = 0$, and in this case the current and the voltage are in phase, the circuit behaving like a purely resistive circuit.

The voltage response and phasor diagrams are shown in Fig. 10.8 and 10.9.

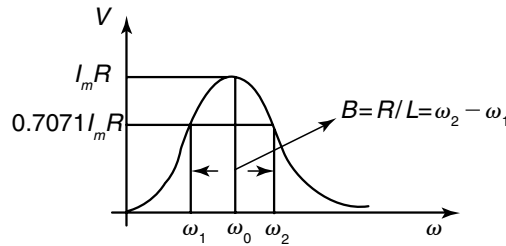


Figure 10.8 Frequency Response of a Parallel – Resonant Circuit

ii Phasor Diagrams

The voltage and current phasor diagrams for an RLC parallel circuit are shown in Fig. 10.9.

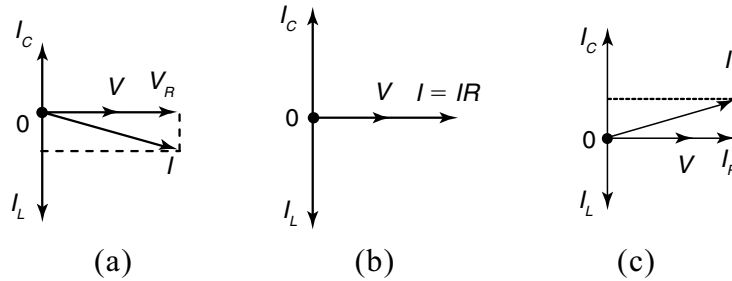


Figure 10.9 Phasor Diagrams (a) $f < f_0$ (b) $f = f_0$ (c) $f > f_0$

i. Bandwidth

We define the half-power bandwidth of the RLC circuit as the range of frequencies (or the width of the frequency band) for which the power dissipated in R is greater than or equal to half the maximum power.

We know that the average power is,

$$P = \frac{|V|^2}{R} \quad \text{where} \quad |V| = \frac{V_m}{\sqrt{2}} = \frac{I_m R}{\sqrt{2}} \quad (22)$$

Maximum power will be, $P_{\max} = \frac{V_m^2}{R}$

Thus, the half-power points occur when

$$P = \frac{P_{\max}}{2} \quad \text{or} \quad |V| = \frac{V_m}{\sqrt{2}}$$

At resonance, the circuit is purely resistive, so that

$$V_m = I_m R \quad (10.23)$$

Therefore, at half-power points,

$$|V| = \frac{V_m}{\sqrt{2}} = \frac{I_m R}{\sqrt{2}} \quad (10.24)$$

From equations (10.19) and (10.24), we get,

$$\frac{I_m}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}} = \frac{I_m R}{\sqrt{2}}$$

Solve for the frequencies, in similar way as for the series circuit, we get,

$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} \quad (10.25)$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} \quad (10.26)$$

By definition, the Bandwidth (BW) is given by,

$$BW = \omega_2 - \omega_1 = \frac{1}{RC} \quad (10.27)$$

Also, from equations (10.25) and (10.26), we get

$$\omega_1 \omega_2 = \frac{1}{LC} = \omega_0^2 \Rightarrow \omega_0 = \sqrt{\omega_1 \omega_2} \quad (10.28)$$

Thus, the resonant frequency is the geometric mean of the half-power frequencies.

Quality Factor or Circuit Magnification Factor (Q)

Here,
$$Q = \frac{\omega_0}{\omega_2 - \omega_1} = \frac{1/\sqrt{LC}}{1/RC} = \omega_0 RC = \frac{R}{\omega_0 L} = R \sqrt{\frac{C}{L}} \quad (10.29)$$

From (10.12) and (10.29), we see that,

$$Q_{\text{series}} = \frac{1}{Q_{\text{parallel}}}.$$

Therefore, for parallel RLC circuit, a high value of Q is achieved via a large value of R resulting high losses. The variations of magnitude and phase angle of voltage in RLC parallel circuit for different values of Quality factor (Q) are shown in Fig. 10.10.

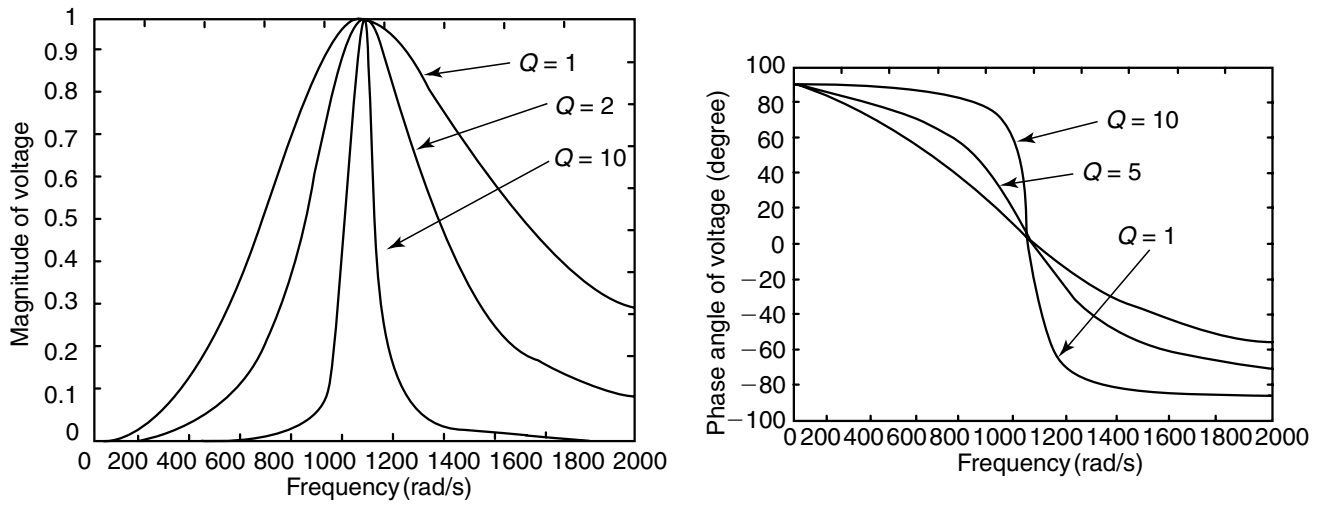


Figure 10.10 Variations of magnitude and phase angle of voltage in RLC parallel circuit for different values of Q

iii. Current Through Elements

Current through Resistance

Since $I = \frac{V_m}{R}$ and at resonance $I_R = I_m$, therefore,

$$V_m = I_m R \quad (10.30)$$

Current through Inductance

We know, $I_L = \frac{V_m}{\omega L}$, using equation (10.19), we get,

$$|I_L| = \frac{|V|}{\omega L} = \frac{I_m}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}} \frac{1}{\omega L}$$

At resonance, (i.e. $\omega_0 L = \frac{1}{\omega_0 C}$),

$$|I_L| = \frac{RI_m}{\omega_0 L} = QI_m \quad (10.31)$$

Current through Capacitance

We know, $I_C = Vj\omega C$; using equation (10.19), we get,

$$|I_C| = |V| \omega C = \frac{I_m}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}} \omega C$$

At resonance, (i.e. $\omega_0 L = \frac{1}{\omega_0 C}$),

$$|I_C| = I_m \omega_0 RC = QI_m \quad (10.32)$$

From equations (10.31) and (10.32) we see that both I_L and I_C may be very large at resonance and they will add to zero (current through L and C are 180° out of phase) so that the entire current will be flowing through the resistance. For this reason, the resonance in parallel RLC circuit is known as *current resonance*. Also, at resonant condition, from equations (10.31) and (10.32) we get,

$$|I_C| = |I_L| = QI_m \Rightarrow Q = \frac{|I_C|}{I_m} = \frac{|I_L|}{I_m}$$

Thus, quality factor for parallel circuit is also defined as the ratio of current through the inductor or capacitor at resonance to the source current. For this reason, Q is also known as *circuit magnification factor* (here, current magnification).

iv. Impedance at Frequencies near Resonant Frequency

The impedance of the RLC parallel circuit at any frequency is given by,

$$\begin{aligned} Z &= \frac{1}{\frac{1}{R} + \frac{1}{j\omega L} + j\omega C} \\ &= \frac{R}{1 + \frac{R}{j\omega L} + j\omega RC} \\ &= \frac{R}{1 + j\left(\omega RC - \frac{R}{\omega L}\right)} \\ &= \frac{R}{1 + j\left(\omega_0 RC \frac{\omega}{\omega_0} - \frac{R}{\omega_0 L} \frac{\omega_0}{\omega}\right)} \\ Z &= \frac{R}{1 + jQ\left(1 + \delta - \frac{1}{1 + \delta}\right)} \\ &= \frac{R}{1 + jQ\delta\left(\frac{2 + \delta}{1 + \delta}\right)} \end{aligned}$$

At frequencies near the resonant frequency, $\delta \ll 1$. $\therefore (2 + \delta) \approx 2$ and $(1 + \delta) \approx 1$.

$\therefore$ Impedance near resonant frequency,

$$Z = \frac{R}{1 + jQ2\delta}$$

Note: (i) At $\omega = \omega_2$ (half-power frequency) $2Q\delta = 1$; $Z = \frac{R}{1+j}$

(ii) At $\omega = \omega_1$ (half-power frequency) $2Q\delta = -1$; $Z = \frac{R}{1-j}$

10.4 RELATION BETWEEN DAMPING RATIO AND QUALITY FACTOR

The frequency response may be related to the natural response as discussed below.
From chapter 5, it was seen that the damping ratio is,

$$\alpha = \xi\omega_0 = \frac{R}{2L}; \text{ for series RLC circuit}$$

$$= \frac{1}{2RC}; \text{ for parallel RLC circuit}$$

From equation (10.29),

$$Q = \frac{\omega_0}{\omega_2 - \omega_1} = \omega_0 RC = \frac{\omega_0}{2\alpha}$$

$$\Rightarrow \alpha = \frac{\omega_2 - \omega_1}{2} = \frac{\omega_0}{2Q} \quad (10.33)$$

Also, from chapter 5, the damped frequency of oscillation is,

$$\omega_d = \omega_0 \sqrt{1 - \xi^2} = \sqrt{\omega_0^2 - \alpha^2} = \omega_0 \sqrt{1 - \frac{1}{4Q^2}} \quad (10.34)$$

From equation (10.34), it is seen that the damping co-efficient α is inversely proportional to the quality factor, Q . Also, the transition from underdamped to overdamped response occurs when $\omega_0^2 = \alpha^2$ (i.e. at the critical damping condition). Thus, from equation (10.33), we see that the *condition for critical damping* is,

$$Q = \frac{1}{2}$$

Similarly, from equation (10.33),

When $\alpha^2 > \omega^2$, $Q < \frac{1}{2}$ (overdamped)

and

When $\alpha^2 < \omega^2$, $Q > \frac{1}{2}$ (underdamped)

10.5 A MORE REALISTIC PARALLEL RESONANT CIRCUIT

A more realistic parallel-resonant circuit is shown in Fig. 10.10. It is a more realistic model because it accounts for the losses in the inductor through resistance R_L and losses in the capacitor through resistance R_C .

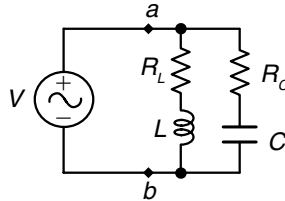


Figure 10.11 Real Parallel Resonant Circuit

Here,

$$Z_1 = (R_L + j\omega L)$$

$$Z_2 = \left(R_C + \frac{1}{j\omega C} \right)$$

$$\begin{aligned} Y &= \frac{1}{R_L + j\omega L} + \frac{j\omega C}{1 + j\omega C R_C} = \frac{R_L - j\omega L}{R_L^2 + \omega^2 L^2} + \frac{j\omega C(1 - j\omega C R_C)}{1 + \omega^2 C^2 R_C^2} \\ &= \left(\frac{R_L}{R_L^2 + \omega^2 L^2} + \frac{\omega^2 C^2 R_C}{1 + \omega^2 C^2 R_C^2} \right) + j \left(\frac{\omega C}{1 + \omega^2 C^2 R_C^2} - \frac{\omega L}{R_L^2 + \omega^2 L^2} \right) \quad (10.35) \end{aligned}$$

For resonance to occur, the imaginary part of the admittance should be zero.

$$\therefore \frac{\omega C}{1 + \omega^2 C^2 R_C^2} = \frac{\omega L}{R_L^2 + \omega^2 L^2}$$

$$\Rightarrow R_L^2 C + \omega^2 L^2 C^2 = L + \omega^2 L R_C^2 C^2$$

$$\Rightarrow \omega^2 L C (R_C^2 - L) = R_L^2 C - L$$

$$\Rightarrow \omega_0 = \sqrt{\frac{1}{LC} \left(\frac{L - C R_L^2}{L - C R_C^2} \right)}$$

Thus, the resonant frequency is given by,

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} \left(\frac{L - C R_L^2}{L - C R_C^2} \right)}$$

We consider the following four conditions:

Case (1): $R_C = 0$

Here, the resonant frequency is,

$$\omega_0 = \sqrt{\frac{1}{LC} \left(1 - \frac{CR_L^2}{L} \right)}$$

Under this condition, the total admittance becomes, from equation (10.34)

$$Y_0 = \frac{R_L}{R_L^2 + \omega_0^2 L^2}$$

From resonant frequency, $\omega_0^2 = \frac{1}{LC} - \frac{R_L^2}{L^2} \Rightarrow (R_L^2 + \omega_0^2 L^2) = \frac{L}{C}$

Therefore, the admittance is, $Y_0 = \frac{R_L}{L/C}$ and the impedance under resonant condition becomes,

$$Z_0 = \frac{L}{CR_L} = \frac{1}{R_L} (R_L^2 + \omega_0^2 L^2) = R_L \left(1 + \frac{\omega_0^2 L^2}{R_L^2} \right) = R_L (1 + Q^2)$$

This impedance is known as *Dynamic Resistance of the parallel tuned circuit*. It is seen that lower the value of resistance R_L , higher is the value of dynamic resistance of the parallel circuit. The current drawn from the supply at resonance is,

$$I_0 = V \frac{CR_L}{L}$$

This current is termed as the *make-up current*.

The current in the capacitor or inductor branch is called *forced oscillatory current*, given by,

$$I = V\omega_0 C$$

This type of circuit is known as a *rejector circuit*, since its impedance approaches a maxima and, therefore, the resultant current is a minima at (or near) resonant frequency.

Impedance near resonant frequency

It is obtained by the procedures as explained below.

Impedance at any frequency, ω is,

$$\begin{aligned} Z &= \frac{(R_L + j\omega L) \left(-j/\omega C \right)}{R_L + j \left(\omega L - \frac{1}{\omega C} \right)} \\ &= \frac{R_L \left(1 + \frac{j\omega L}{R_L} \right) \left(-j/\omega C \right)}{R_L \left[1 + \frac{j\omega L}{R_L} \left(1 - \frac{1}{\omega^2 LC} \right) \right]} \end{aligned}$$

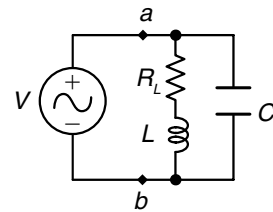


Figure 10.12 Parallel Tuned Resonant Circuit with Lossless Capacitor

$$= \frac{\frac{L}{CR_L} \left(1 - j \frac{R_L}{\omega L} \right)}{1 + j \frac{\omega L}{R_L} \left(1 - \frac{1}{\omega^2 LC} \right)} \quad (10.36)$$

Now, $\frac{\omega L}{R_L} = \frac{\omega_0 L}{R_L} \times \frac{\omega}{\omega_0} = Q \frac{\omega}{\omega_0} = Q(1 + \delta) \quad \{ \delta = \frac{\omega - \omega_0}{\omega_0} \}$

and, $\frac{1}{\omega^2 LC} = \frac{\omega_0^2}{\omega^2} \times \frac{1}{\omega_0^2 LC} = \frac{1}{(1 + \delta)^2} \times 1 = \frac{1}{(1 + \delta)^2}$

{As, for high values of Q ($Q > 10$), $\therefore \omega_0^2 = \frac{1}{LC} \Rightarrow \frac{1}{\omega_0^2 LC} = 1$ }

Substituting these values in (10.36),

$$\begin{aligned} Z &= \frac{\frac{L}{CR_L} \left(1 - j \frac{R_L}{\omega L} \right)}{1 + j \frac{\omega L}{R_L} \left(1 - \frac{1}{\omega^2 LC} \right)} \\ &= \frac{Q^2 R_L \left[1 - \frac{j}{Q(1 + \delta)} \right]}{1 + jQ\delta(1 + \delta) \left[\frac{2 + \delta}{(1 + \delta)^2} \right]} \\ &= \frac{Q^2 R_L \left[1 - \frac{j}{Q(1 + \delta)} \right]}{1 + jQ\delta \left[\frac{2 + \delta}{1 + \delta} \right]} \end{aligned}$$

Near resonant frequency, $(2 + \delta) \approx 2$ and $(1 + \delta \approx 1)$

$\therefore$ Impedance near resonant frequency,

$$Z_0 = \frac{Q^2 R_L \left[1 - \frac{j}{Q} \right]}{1 + j2Q\delta}$$

NB:

At resonant frequency, $\delta = 0$,

$\therefore$ Impedance at resonance frequency, $Z_0 = Q^2 R_L \left[1 - \frac{j}{Q} \right]$

(i) For large values of Q (i.e., $Q > 10$), at resonant frequency, the impedance is,

$$Z_0 \approx Q^2 R_L = \frac{L}{CR_L}$$

(ii) For large values of Q (i.e., $Q > 10$), for near resonant frequency, the impedance is,

$$Z_0 = \frac{Q^2 R_L}{1 + j2Q\delta} = \frac{R_d}{1 + j2Q\delta} \quad [\text{as, Dynamic Resistance, } R_d = \frac{L}{CR_L}]$$

$$\therefore \quad \boxed{\frac{Z_0}{R_d} = \frac{1}{1 + j2Q\delta}}$$

Current in Parallel Tuned Circuit

Let,

Current delivered by the source = I_s ,

Current in the inductor branch = I_L ,

Current in the capacitor branch = I_C ,

$\therefore$ Power delivered by the source, $P_s = I_s^2 R_d$ [R_d = Dynamic Resistance]

Power dissipated in the parallel circuit, $P = I_L^2 R_L$

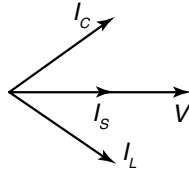


Figure 10.13 Phasor diagram for Parallel Tune Resonant Circuit with Lossless Capacitor

Now, $P_s = P$

$$\Rightarrow I_s^2 R_d = I_L^2 R_L \quad (10.37)$$

$$\Rightarrow I_L^2 = I_s^2 \frac{R_d}{R_L} = I_s^2 \frac{L}{CR_L^2} \left[\because R_d = \frac{L}{CR_L} \right]$$

$$\Rightarrow I_L = I_s \frac{1}{R_L} \sqrt{\frac{L}{C}}$$

Thus, the parallel tuned circuit is a current amplifier.

Voltage across the capacitor, $V_C = I_s R_d$

and current through the capacitor, $I_C = \omega_0 C V_C$

$$\therefore I_s = \frac{V_C}{R_d} = \frac{I_C}{\omega_0 C R_d}$$

From (10.37),

$$I_s^2 R_d = I_L^2 R_L$$

$$\begin{aligned}
 \Rightarrow \quad & \frac{I_s^2}{I_L^2} = \frac{R_L}{R_d} \\
 \Rightarrow \quad & \frac{I_C^2}{\omega_0^2 C^2 R_d^2 I_L^2} = \frac{R_L}{R_d} \\
 \Rightarrow \quad & \frac{I_C^2}{I_L^2} = \frac{R_L}{R_d} \times \omega_0^2 C^2 R_d^2 = \omega_0^2 LC \left[\because R_d = \frac{L}{CR_L} \right] \\
 \Rightarrow \quad & \frac{I_C}{I_L} = \sqrt{LC} \omega_0 = \sqrt{LC} \left[\sqrt{\frac{1}{LC} \left(1 - \frac{CR_L^2}{L} \right)} \right] = \sqrt{\left(1 - \frac{CR_L^2}{L} \right)} \\
 & = \sqrt{1 - \frac{1}{Q^2}} \left[\because Q = \frac{1}{R_L} \sqrt{\frac{L}{C}} \right] \\
 \therefore \quad & \boxed{\frac{I_C}{I_L} = \sqrt{1 - \frac{1}{Q^2}}}
 \end{aligned}$$

If R_L is very large, the currents are not equal. However, for very low value of R_L (for very high value of Q , $Q \geq 10$), both the currents are equal. Higher the value of Q , higher will be I_C and I_L and lower will be the source current, I_s . As $Q \rightarrow \infty$, both I_C and I_L tend to infinity and I_s tends to zero.

Effect of Source Resistance over Bandwidth and Quality Factor

Bandwidth for the parallel tuned circuit is affected by the source resistance as explained below

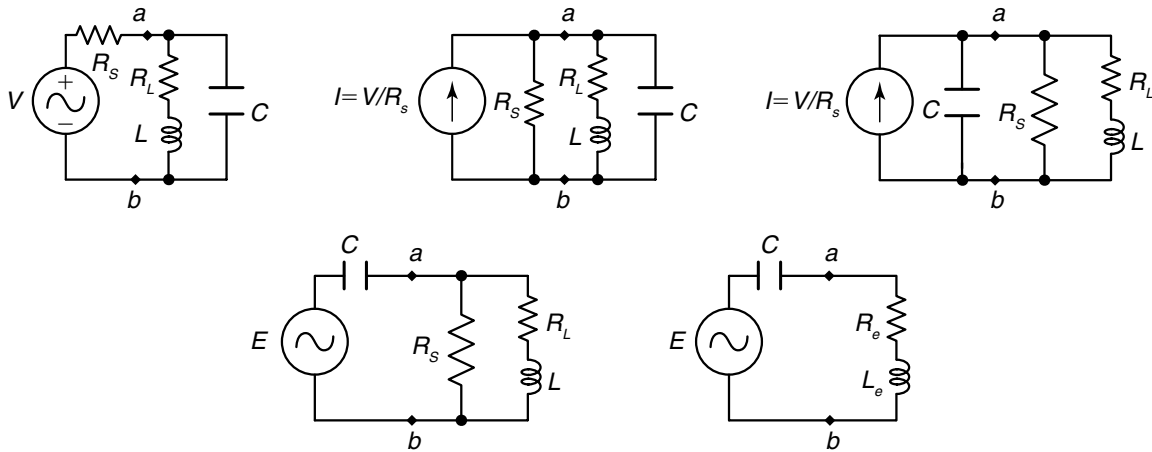


Figure 10.14 Simplification of Real Parallel Circuit to get Series Equivalent Circuit

By circuit reduction methods, the parallel tuned circuit is simplified as shown from Fig. 10.14 (i) to Fig. 10.14 (v),

$$Z_e = \frac{(R_L + j\omega L) R_s}{(R_L + j\omega L) + R_s}$$

$$\begin{aligned}
&= \frac{R_L R_s + j\omega L R_s}{(R_L + j\omega L) + R_s} \\
&= \frac{(R_L R_s + j\omega L R_s) \{(R_L + R_s) - j\omega L\}}{\{(R_L + R_s) + j\omega L\} \{(R_L + R_s) - j\omega L\}} \\
&= \frac{\{R_L R_s (R_L + R_s) + \omega^2 L^2 R_s\} + j\{\omega L R_s (R_L + R_s) - \omega L R_s\}}{(R_L + R_s)^2 + \omega^2 L^2} \\
&= R_e + j\omega L_e
\end{aligned}$$

Therefore, the equivalent resistance is,

$$R_e = \frac{R_L R_s (R_L + R_s) + \omega^2 L^2 R_s}{(R_L + R_s)^2 + \omega^2 L^2}$$

and the equivalent inductance is,

$$L_e = \frac{L R_s^2}{(R_L + R_s)^2 + \omega^2 L^2}$$

Since the equivalent circuit is an RLC series circuit, the effective quality factor is,

$$\begin{aligned}
Q &= \frac{\omega L_e}{R_e} = \frac{\omega L R_s}{R_L (R_L + R_s) + \omega^2 L^2} = \frac{1}{\frac{R_L}{\omega L} + \frac{R_L^2 + \omega^2 L^2}{\omega L R_s}} = \frac{1}{\frac{1}{Q} + \left(\frac{R_L^2 + \omega^2 L^2}{R_L \omega L R_s} \right) R_L} \\
&\quad \left(\because Q = \frac{\omega L}{R_L} \right) \\
&= \frac{1}{\frac{1}{Q} + \left(\frac{R_L}{\omega L} \right) \frac{R_d}{R_s}} = \frac{1}{\frac{1}{Q} + \frac{1}{Q} \left(\frac{R_d}{R_s} \right)} = \frac{Q}{1 + \frac{R_d}{R_s}} \\
&\quad \left(\text{as, at resonance } R_L^2 + \omega^2 L^2 = \frac{L}{C} \text{ and } R_d = \frac{L}{C R_L} \right)
\end{aligned}$$

Therefore,

$$Q_e = \frac{\omega L / R_L}{\left(1 + \frac{L}{C R_L R_s} \right)} = \frac{Q}{\left(1 + \frac{R_d}{R_s} \right)}$$

Therefore, the effective bandwidth will be,

$$(BW)_e = \frac{\omega_0}{Q_e} = \frac{1}{Q} \left(1 + \frac{R_d}{R_s} \right) \omega_0$$

$$\therefore (BW)_e = \frac{\omega_0}{Q} \left(1 + \frac{R_d}{R_s} \right) = \frac{\omega_0}{Q} \left[1 + \frac{L}{CR_L R_s} \right]$$

Thus, the bandwidth of the circuit depends upon the circuit constants (R_L , L and C) and the source resistance (R_s). For a given resonant frequency, the circuit will be less selective and bandwidth will be large if L is large and C is small. For more selectivity, the value of L should be reduced and C increased; however, this may reduce the value of R_d , which is undesirable.

NB: If the source resistance is infinite ($R_s \rightarrow \infty$), then the bandwidth and quality factor of this parallel tuned circuit is the same as those of the RLC series circuit.

To obtain maximum possible value of power delivered from the source to the load, we have,

$$R_s = R_d$$

and under this condition, the bandwidth becomes,

$$BW = \frac{2}{Q} \omega_0$$

Example 10.4 A coil of inductance 1 H and 10 Ω resistance is connected in parallel with 100 μF capacitor. If the supply voltage is 200 V, find the resonant frequency and the current at resonance.

Sol:

Here, $R_L = 10 \Omega$, $L = 1 \text{ H}$, $C = 100 \mu\text{F}$, $V = 200 \text{ V}$

Resonant frequency is,

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}} = \frac{1}{2\pi} \sqrt{\frac{1}{1 \times 100 \times 10^{-6}} - \frac{10^2}{1^2}} = 15.84 \text{ Hz} \quad \text{Ans}$$

At resonance, the impedance of the circuit is,

$$Z_0 = \frac{L}{CR_L} = \frac{1}{100 \times 10^{-6} \times 10} = 1000 \Omega$$

$\therefore$ Current in the circuit at resonance is, $I_0 = \frac{V}{Z_0} = \frac{200}{1000} = 0.2 \text{ A} \quad \text{Ans}$

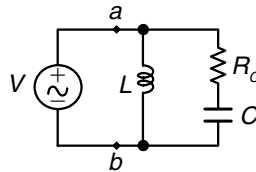


Figure 10.15 Real Parallel Resonant Circuit with Lossless Inductor

Case (2): $R_L = 0$:

Here, the resonant frequency is,

$$\omega_0 = \sqrt{\frac{1}{LC} \left(\frac{L}{L - CR_C^2} \right)}$$

Under this condition, the total admittance becomes, from equation (10.35)

$$Y_0 = \frac{\omega_0^2 C^2 R_C}{1 + \omega_0^2 C^2 R_C^2}$$

From resonant frequency,

$$\omega_0^2 = \frac{1}{C(L - CR_C^2)} \Rightarrow (1 + \omega_0^2 C^2 R_C^2) = \omega_0^2 LC$$

Therefore, the admittance is, $Y_0 = \frac{\omega_0^2 C^2 R_C}{\omega_0^2 LC} = \frac{CR_C}{L}$ and the impedance under resonant condition becomes,

$$Z_0 = \frac{L}{CR_C}$$

This is the *Dynamic Resistance of the parallel tuned circuit*. Here also, lower the value of resistance R_C , higher is the value of dynamic resistance of the parallel circuit. The *make-up current* i.e., current drawn from the supply at resonance is,

$$I_0 = V \frac{CR_C}{L}$$

and the *forced oscillatory current*, given by, $I = \frac{V}{\omega_0 L}$

This circuit is also a *rejector circuit* as the impedance approaches a maxima and the current a minima.

Approaching in the way similar to case (1), we get the equivalent quality factor and bandwidth of this resonating circuit as,

$$Q_e = \frac{1}{\omega CR_C \left(1 + \frac{R_d}{R_s} \right)} = \frac{Q}{\left(1 + \frac{R_d}{R_s} \right)} \quad \text{where, } Q = \frac{1}{\omega CR_C}$$

and

$$BW = \frac{1}{Q} \left(1 + \frac{R_d}{R_s} \right) \omega_0 = \frac{1}{Q} \left[1 + \frac{L}{CR_C R_s} \right] \omega_0$$

Case (3): $R_L = R_C = 0$:

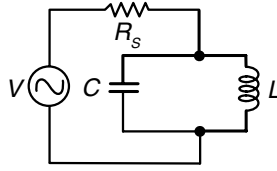


Figure 10.16 Tuned Tank Circuit

Here, the resonant frequency is, $\omega_0 = \sqrt{\frac{1}{LC}}$. This circuit with L and C in parallel is termed as *tuned tank circuit*. At resonance, with capacitive and inductive reactances equal to each other, the total impedance increases to infinity, meaning that the tank circuit draws no current from the AC power source. However, for this case, there will be some circulating current, given by,

$$I = I_C = V\omega C = V \frac{1}{\sqrt{LC}} C \quad (\text{if } R_s \ll)$$

$$\Rightarrow I = V \sqrt{\frac{C}{L}}$$

The quality factor and bandwidth of this circuit will be same as those of a simple parallel RLC circuit; i.e.,

$$Q = \frac{R_s}{\omega L} = \omega C R_s \quad \text{and} \quad BW = \frac{1}{R_s C}.$$

Case (4): $R_L = R_C = \sqrt{\frac{L}{C}}$

$$\text{Here, } R_L = R_C = \sqrt{\frac{L}{C}} = \sqrt{\frac{\omega L}{\omega C}} = \sqrt{X_L X_C}$$

Under this condition, the imaginary part of the admittance is,

$$\begin{aligned} IM(Y) &= \frac{\omega C}{1 + \omega^2 C^2 R_C^2} - \frac{\omega L}{R_L^2 + \omega^2 L^2} \\ &= \frac{X_C}{X_C^2 + R_C^2} - \frac{X_L}{X_L^2 + R_L^2} \\ &= \frac{R_L^2 X_C + X_L^2 X_C - X_L X_C^2 + R_C^2 X_L}{(X_C^2 + R_C^2)(X_L^2 + R_L^2)} \end{aligned}$$

$$IM(Y) = \frac{X_L^2 X_C + X_L^2 X_C - X_L X_C^2 + X_L^2 X_C}{(X_C^2 + R_C^2)(X_L^2 + R_L^2)} = 0$$

Thus, the resonance under this condition will occur at all frequencies.

Example 10.5 Find C which results in resonance in the circuit shown in Fig. 10.17 when $\omega = 5000$ rad/s.

Sol:

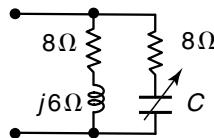


Figure 10.17 Circuit of Example 10.5

Let the capacitive reactance be X_C .

$$\begin{aligned} Y &= \frac{1}{8 + j6} + \frac{1}{8 - jX_C} = \frac{8 - j6}{100} + \frac{8 - jX_C}{64 + X_C^2} \\ &= \left(\frac{8}{100} + \frac{8}{64 + X_C^2} \right) + j \left(\frac{X_C}{64 + X_C^2} - \frac{6}{100} \right) \end{aligned}$$

For resonance to occur, the imaginary part of the admittance should be zero.

$$\therefore \left(\frac{X_C}{64 + X_C^2} - \frac{6}{100} \right) = 0$$

$$\Rightarrow \frac{X_C}{64 + X_C^2} = \frac{6}{100}$$

$$\Rightarrow 6X_C^2 - 100X_C + 384 = 0$$

$$\Rightarrow X_C = 10.67 \, \Omega \text{ or } 6 \, \Omega$$

$$\therefore X_C = \frac{1}{\omega C}$$

$$\therefore C = \frac{1}{\omega X_C} = \frac{1}{5000 \times 10.67} = 18.75 \, \mu\text{F} \text{ or } \frac{1}{5000 \times 6} = 33.33 \, \mu\text{F} \text{ Ans}$$

10.6 COMPARISON OF SERIES AND PARALLEL RESONANCE

Difference	Particulars	Series Resonance	Parallel Resonance
	Circuit Impedance at resonance	Minimum	Maximum
	at resonance		
	Circuit Admittance at resonance	Maximum	Minimum
	Current in the circuit at resonance	Maximum	Minimum
	Circuit for $f > f_0$	Inductive	Capacitive
	Circuit for $f < f_0$	Capacitive	Inductive
	Amplification of	Voltage	Current
	Q -factor	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$ or $\frac{1}{R} \sqrt{\frac{L}{C}}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$ or $R \sqrt{\frac{C}{L}}$
Similarities	Power factor of the circuit at resonance	Unity	Unity
	Resonant frequency	$\omega_0 = \frac{1}{\sqrt{LC}}$	$\omega_0 = \frac{1}{\sqrt{LC}}$
	Bandwidth	$BW = \frac{f_0}{Q}$	$BW = \frac{f_0}{Q}$
	Half-power frequencies	$\omega_0 = \sqrt{\omega_1 \omega_2}$	$\omega_0 = \sqrt{\omega_1 \omega_2}$

10.7 UNIVERSAL RESONANCE CURVE

From the expressions of impedance near resonant frequency,
For series RLC circuit,

$$Z = R(1 + j2\delta Q)$$

$$Y = \frac{1}{R(1 + j2\delta Q)}$$

$$\frac{Y}{Y_0} = \frac{1}{1 + j2\delta Q} \quad (10.38)$$

$$\left(\text{where, } Y_0 = \frac{1}{R} = \text{admittance at resonance} \right)$$

For parallel RLC circuit,

$$Z = \frac{R}{1 + j2Q\delta}$$

or, $\frac{Z}{R} = \frac{1}{1 + j2\delta Q}$

or, $\frac{Z}{Z_0} = \frac{1}{1 + j2\delta Q}$ (10.39)

(where, $Z_0 = R = \text{impedance at resonance}$)

For parallel tuned circuit for high values of Q ,

$$Z = \frac{R_d}{1 + j2Q\delta}$$

or, $\frac{Z}{Z_0} = \frac{1}{1 + j2Q\delta}$ (10.40)

(where, $Z_0 = R_d = \text{Dynamic resistance} = \text{impedance at resonance}$)

From (10.38), (10.39) and (10.40), it is seen that the variations of $\frac{Y}{Y_0}$ or $\frac{Z}{Z_0}$ for high Q circuits near resonant frequency are identical for series and parallel circuit, given as,

$$H = \frac{1}{1 + j2\delta Q} = |H| \angle \phi$$

where, $|H| = \frac{1}{\sqrt{1 + 4\delta^2 Q^2}}$ and $\phi = -\tan^{-1}(2\delta Q)$

The variation of magnitude and phase of this are shown in Fig.10.18. These curves are known as *universal resonance curve*. It gives the magnitude and phase of the quantity by which the maximum admittance or impedance of a series or parallel circuit is to be multiplied to obtain the associated admittance or impedance near resonant frequency.

NB: At half-power points, $2\delta Q = \pm 1$, $\therefore |H| = \frac{1}{\sqrt{2}} = 0.707$ and $\phi = \pm 45^\circ$.

10.8 APPLICATIONS OF RESONANCE

Resonance is a very valuable property of reactive AC circuits, employed in a variety of applications.

One use for resonance is to establish a condition of stable frequency in circuits designed to produce AC signals. For example, when we tune a radio to a particular station, the LC circuits are set at resonance for that particular carrier frequency. Usually, a parallel (tank) circuit is used for this

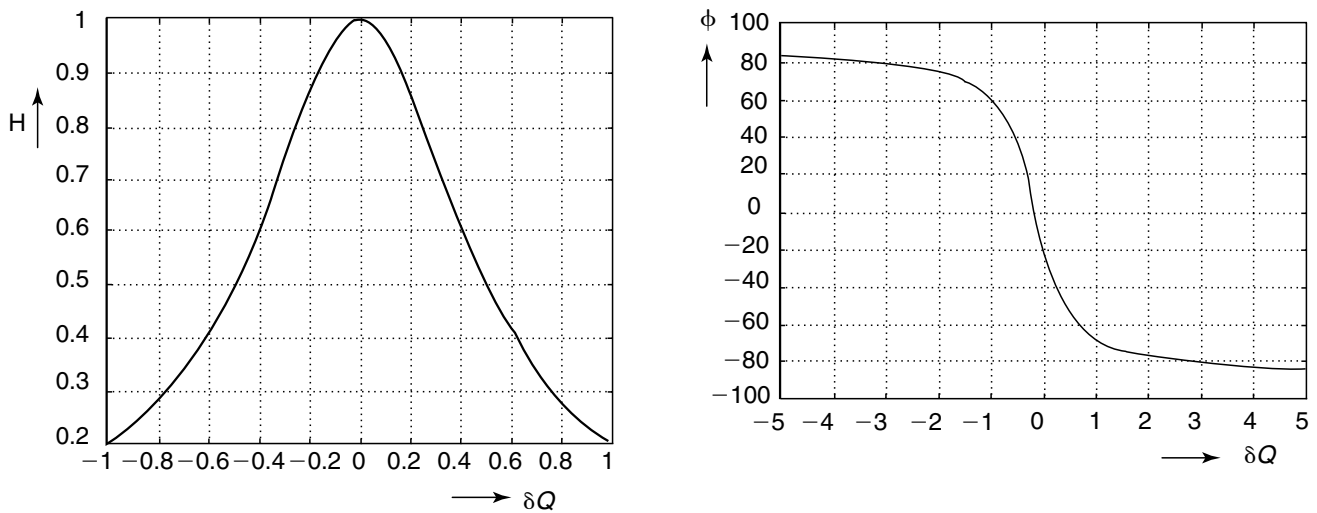


Figure 10.18 *Universal Resonance Curves*

purpose, with the capacitor and inductor directly connected together, exchanging energy between each other. Just as a pendulum can be used to stabilize the frequency of a clock mechanism's oscillations, so can a tank circuit be used to stabilize the electrical frequency of an *AC oscillator circuit*.

Another use for resonance is in applications where the effects of greatly increased or decreased impedance at a particular frequency are desired. A resonant circuit can be used to 'block' (present high impedance toward) a frequency or range of frequencies, thus acting as a 'filter' to strain certain frequencies. In fact, these particular circuits are called *filters*, discussed in the preceding chapter.

A parallel resonant circuit can also be used as load impedance in output circuits of RF amplifiers. Due to high impedance, the gain of amplifier is maximum at resonant frequency.

Therefore, the applications of resonant effects can be summarized as follows:

1. Most common application of resonance is tuning i.e., as an **oscillator circuit**.
2. A series resonant circuit is used as **voltage amplifier**.
3. A parallel resonant circuit is used as **current amplifier**.
4. A resonant circuit is used as **filters**.
5. A realistic parallel resonant circuit is used as **current rejector**.
6. A parallel resonant circuit is used as **load impedance in output circuits of RF amplifiers**.
7. A parallel resonant circuit can be used in **induction heating**.

In designing any mechanical systems or civil structure or electrical system, the effects of resonance must be taken into consideration. Otherwise, the oscillations of the system in certain conditions may be so large that the system may be damaged.

SOLVED PROBLEMS

1. A series *RLC* circuit has the values: $R = 10 \Omega$, $L = 0.01 \text{ H}$, $C = 100 \mu\text{F}$. Calculate resonant frequency, quality factor, bandwidth, and the half-power frequencies.

Sol:

Here, $R = 10 \Omega$, $L = 0.01 \text{ H}$, $C = 100 \mu\text{F}$.

$$\therefore \text{Resonant frequency, } f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.01 \times 100 \times 10^{-6}}} = 159.15 \text{ Hz } \text{Ans.}$$

$$\therefore \text{Quality factor, } Q = \frac{\omega_0 L}{R} = \frac{1}{\sqrt{LC}} \times \frac{L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{10} \sqrt{\frac{0.01}{100 \times 10^{-6}}} = 1 \text{ Ans.}$$

$$\therefore \text{Bandwidth, } BW = \frac{R}{L} = \frac{10}{0.01} = 1,000 \text{ rad/s } \text{Ans.}$$

To find half-power frequencies, we have,

$$(\omega_2 - \omega_1) = \frac{R}{L} = 1000$$

$$\text{and } \omega_1 \omega_2 = \frac{1}{LC} = 10^6$$

$$(\omega_2 + \omega_1) = \sqrt{(\omega_2 - \omega_1)^2 + 4\omega_1 \omega_2} = \sqrt{10^6 + 4 \times 10^6} = 2.36 \times 10^3$$

Adding equations (i) + (ii), $\omega_2 = 1.618 \times 10^3 \text{ rad/s } \text{Ans.}$

Subtracting equations (ii) – (i), $\omega_1 = 0.618 \times 10^3 \text{ rad/s } \text{Ans.}$

2. A series *RLC* circuit has the values: $R = 100 \Omega$, $L = 0.02 \text{ H}$, $C = 0.02 \mu\text{F}$. Calculate frequency of resonance. A variable frequency sinusoidal voltage of value 50V is applied to the circuit. Find the frequency at which voltage across L and C is the maximum. Also calculate voltage across L and C at frequency of resonance. Find the maximum current in the circuit.

Sol:

Here, $R = 100 \Omega$, $L = 0.02 \text{ H}$, $C = 0.02 \mu\text{F}$.

$$\therefore \text{Resonant frequency, } f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.02 \times 0.02 \times 10^{-6}}} = 7.957 \text{ kHz } \text{Ans.}$$

The frequency at which voltage across C is the maximum is given by,

$$f_C = \left(\frac{1}{2\pi} \right) \sqrt{\left(\frac{1}{LC} - \frac{R^2}{2L^2} \right)} = \left(\frac{1}{2\pi} \right) \sqrt{\left(\frac{1}{0.02 \times 0.02 \times 10^{-6}} - \frac{100^2}{2(0.02)^2} \right)} = 7.937 \text{ kHz } \text{Ans}$$

The frequency at which voltage across L is the maximum is given by,

$$f_L = \left(\frac{1}{2\pi} \right) \left(\frac{1}{\sqrt{LC - \frac{C^2 R^2}{2}}} \right)$$

$$= \left(\frac{1}{2\pi} \right) \left(\frac{1}{\sqrt{0.02 \times 0.02 \times 10^{-6} - \frac{(0.02 \times 10^{-6})^2 \times 100^2}{2}}} \right) = 7.977 \text{ kHz } Ans$$

Voltage across the inductance or the capacitance at resonance is,

$$|V_L| = |V_C| = QV_m = \omega_0 \frac{L}{R} V_m = 2\pi f_0 \frac{L}{R} V_m = 2 \times \pi \times 7.957 \times 10^3 \times \frac{0.02}{100} \times 50 = 500 \text{ V } Ans$$

$$\therefore V_L = 500 \angle 90^\circ \text{ V } Ans$$

$$\therefore V_C = 500 \angle -90^\circ \text{ V } Ans$$

The maximum current in the circuit is given by,

$$I_{\max} = \frac{V_m}{R} = \frac{50}{100} = 0.5 \text{ A } Ans$$

3. For a series RLC circuit with $R = 2 \Omega$, $L = 1 \text{ mH}$, $C = 0.4 \mu\text{F}$ and a supply voltage $v(t) = 20 \sin \omega t$, find:

- The resonant frequency (ω_0);
- The half-power frequencies (ω_1 and ω_2);
- The quality factor and bandwidth; and
- The amplitude of the current at ω_0 , ω_1 and ω_2 .

Sol:

Here, $R = 2 \Omega$, $L = 0.001 \text{ H}$, $C = 0.4 \mu\text{F}$

$$(a) \therefore \text{Resonant frequency, } \omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.001 \times 0.4 \times 10^{-6}}} = 50 \text{ krad/s } Ans.$$

(b) To find half-power frequencies, we have,

$$(\omega_2 - \omega_1) = \frac{R}{L} = 2000 \quad (i)$$

$$\text{and } \omega_1 \omega_2 = \frac{1}{LC} = 25 \times 10^6$$

$$\therefore (\omega_2 + \omega_1) = \sqrt{(\omega_2 - \omega_1)^2 + 4\omega_1 \omega_2} = \sqrt{4 \times 10^6 + 4 \times 25 \times 10^6} = 100 \times 10^3 \quad (ii)$$

Adding equations (i) + (ii), $\omega_2 = 51 \text{ krad/s } Ans$

Subtracting equations (ii) - (i), $\omega_1 = 49 \text{ krad/s } Ans$

(c) $\therefore$ Quality factor, $Q = \frac{\omega_0 L}{R} = \frac{1}{\sqrt{LC}} \times \frac{L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{2} \sqrt{\frac{0.001}{0.4 \times 10^{-6}}} = 25 \text{ Ans.}$

$\therefore$ Bandwidth, $BW = \frac{R}{L} = \frac{2}{0.001} = 2 \text{ krad/s Ans.}$

(b) The amplitudes of the currents are,

$$|I|_{\omega=\omega_0} = \frac{V}{R} = \frac{20}{2} = 10 \text{ A Ans.}$$

$$|I|_{\omega=\omega_1=\omega_2} = \frac{|I|}{\sqrt{2}} = \frac{10}{\sqrt{2}} = 7.071 \text{ A Ans.}$$

4. A series-connected circuit has $R = 4 \Omega$ and $L = 25 \text{ mH}$. (a) Calculate the value of C that will produce a quality factor of 50. (b) Find ω_1 , ω_2 and BW. (c) Determine the average power dissipated at $\omega = \omega_0$, ω_1 , ω_2 . Take $V_m = 100 \text{ V}$.

Sol:

Here, $R = 4 \Omega$, $L = 25 \text{ mH} = 0.025 \text{ H}$, $Q = 50$, $V_m = 100 \text{ Volt}$

(a)

$$\therefore Q = \frac{\omega_0 L}{R} = \frac{1}{\sqrt{LC}} \times \frac{L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$\therefore C = \frac{L}{Q^2 R^2} = \frac{0.025}{50^2 \times 4^2} = 0.625 \mu\text{F Ans.}$$

(b)

$$\begin{aligned} \omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = -\frac{4}{2 \times 0.025} + \sqrt{\left(\frac{4}{2 \times 0.025}\right)^2 + \frac{1}{0.025 \times 0.625 \times 10^{-6}}} \\ &= -80 + 8000 \\ &= 7920 \text{ rad/s Ans.} \end{aligned}$$

$$\begin{aligned} \omega_2 &= -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = -\frac{4}{2 \times 0.025} - \sqrt{\left(\frac{4}{2 \times 0.025}\right)^2 + \frac{1}{0.025 \times 0.625 \times 10^{-6}}} \\ &= -80 - 8000 \\ &= 8080 \text{ rad/s Ans.} \end{aligned}$$

(c) Average power dissipated at resonant frequency,

$$|P_{\text{av}}|_{\omega=\omega_0} = \frac{V_{\text{rms}}^2}{R} = \frac{\left(\frac{100}{\sqrt{2}}\right)^2}{2} = 1.250 \text{ kW Ans.}$$

Average power dissipated at half-power frequencies,

$$|P_{av}|_{\omega=\omega_1=\omega_2} = \frac{|P_{av}|_{\omega=\omega_0}}{2} = \frac{1.25}{2} = 0.625 \text{ kW} \quad \text{Ans.}$$

- 5. A series RLC circuit with $R = 10 \, \Omega$, $L = \frac{100}{314} \text{ H}$, $C = \frac{1}{31,400} \text{ F}$ is excited from a 10 V, 50 Hz ($\omega = 314 \text{ rad/s}$) source. Determine the rms values of voltage across (i) resistance, (ii) inductance, (iii) capacitance. Give explanation if any of the answer is more than the source voltage.**

Sol:

Here, $\omega = 314 \text{ rad/s}$, $R = 10 \, \Omega$, $L = \frac{100}{314} \text{ H}$, $C = \frac{1}{31,400} \text{ F}$, $V = 10 \text{ V}$

$$\therefore X_L = \omega L = 314 \times \frac{100}{314} = 100 \, \Omega \quad \text{and} \quad X_C = \frac{1}{\omega C} = \frac{1}{314 \times \frac{1}{31,400}} = 100 \, \Omega$$

Since, $X_L = X_C$, the circuit is under resonance.

At resonance, the current in the circuit, $I_0 = \frac{V}{R} = \frac{10}{10} = 1 \text{ A}$

- (i) $\therefore$ Voltage across resistance, $V_R = I_0 \times R = 1 \times 10 = 10 \text{ V} \quad \text{Ans}$
 (ii) $\therefore$ Voltage across inductance, $V_L = I_0 \times \omega L = 1 \times 100 = 100 \text{ V} \quad \text{Ans}$
 (iii) $\therefore$ Voltage across capacitance, $V_C = I_0 \times \frac{1}{\omega C} = 1 \times 100 = 100 \text{ V} \quad \text{Ans}$

At resonance, the voltage drops across the inductance and the capacitance will be equal in magnitude but opposite in phase and thus will nullify each other so that the supply voltage will be equal to the voltage drop across the resistance.

- 6. A $20 \, \Omega$ resistor is connected in series with an inductor, a capacitor and an ammeter across a 25 V variable frequency supply. When the frequency is 400 Hz, the current is at its maximum value of 0.5 A and the potential difference across the capacitor is 150 V. Calculate:**

- The capacitance of the capacitor.
- The resistance and inductance of the inductor.

Sol:

Here, $R = 20 \, \Omega$, $V_m = 25 \text{ Volt}$, $f = 400 \text{ Hz}$, $I = 0.5 \text{ A}$, $V_C = 150 \text{ Volt}$

When the current is the maximum, the circuit is in resonance and hence total reactances

$$X_L \sim X_C = 0$$

- The capacitance value is calculated as,

$$\therefore X_C = \frac{V_C}{I} = \frac{150}{0.5} = 300 \, \Omega$$

$$\Rightarrow \frac{1}{\omega C} = 300 \Rightarrow C = \frac{1}{2\pi \times 400 \times 300} = 1.325 \mu\text{F} \text{ Ans.}$$

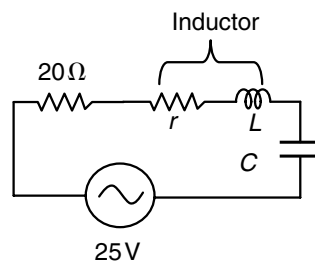
$$\text{ii. } \because X_L \sim X_C = 0,$$

$$\therefore X_L = X_C$$

$$\Rightarrow \omega L = 300 \Omega$$

$$\Rightarrow L = \frac{300}{2\pi \times 400} = 0.119 \text{ H} \text{ Ans.}$$

Also, at resonance, circuit resistance = circuit impedance



Let, r = resistance of the inductor

$$\text{Then } (20 + r) = \frac{25}{0.5} = 50 \Rightarrow r = 30 \Omega \text{ Ans.}$$

7. Voltages across resistance, inductance and capacitance connected in series are 3 V, 4 V and 5 V respectively. If supply voltage has 50 Hz frequency, what is the magnitude of supply voltage? Find the resonant frequency of this series RLC circuit.

Sol:

$$\text{Here, } V_R = 3 \text{ V; } V_L = 4 \text{ V; } V_C = 5 \text{ V, } f = 50 \text{ Hz}$$

$$\text{Supply voltage is, } V = \sqrt{V_R^2 + (V_C - V_L)^2} = \sqrt{3^2 + (5 - 4)^2} = \sqrt{10} = 3.162 \text{ V} \text{ Ans}$$

Now, voltage drop across inductance is,

$$V_L = \omega L \times I \quad (i)$$

Voltage drop across capacitance is,

$$V_C = \frac{1}{\omega C} \times I \quad (ii)$$

By (ii) $\div$ (i), we get,

$$\frac{1}{\omega^2 LC} = \frac{V_C}{V_L} \Rightarrow \frac{1}{LC} = \omega^2 \times \frac{V_C}{V_L} = (2\pi \times 50)^2 \times \frac{5}{4} = 123.37 \times 10^3$$

$\therefore$ Resonant frequency is,

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC}} = \frac{1}{2\pi} \sqrt{123.37 \times 10^3} = 55.9 \text{ Hz} \text{ Ans}$$